

WEEK 5 POWER BI PROJECT REPORT

Healthcare Data Cleaning & Data Modeling

Dataset: Healthcare Dataset

Tool: Microsoft Power BI

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1. Project Overview

The Week 5 task focuses on preparing a healthcare dataset for analysis by cleaning the raw records, standardizing categorical fields, creating derived patient attributes, building dimension and fact tables, and developing an interactive Power BI report.

Problem Statement

A healthcare organization wants to analyze patient information, medical conditions, admissions, and healthcare-related characteristics. The raw data must be cleaned and modeled before creating an interactive Power BI report.

Objectives

- Import and inspect patient records, admission types, and billing fields in Power Query.
- Handle null or inconsistent values and standardize categorical text attributes.
- Create derived conditional columns such as Age Group.
- Structure dimension tables (Dim_Patient and Dim_Admission) and a fact table (Billing).
- Create a star-schema style model and define relationships between the tables.
- Create DAX measures for patient, billing, length-of-stay and readmission analysis.
- Build an interactive dashboard with slicers, summary cards, tables and charts.

Tasks and Deliverables

- Group patient ages into logical demographic categories: 0–18, 19–35, 36–60 and 60+.
- Standardize admission types: Emergency, Elective and Urgent.
- Set up relationships between patient/admission dimensions and the billing fact table.
- Output a cleaned data model ready for healthcare analytics.

2. Data Cleaning and Preparation

The healthcare data was prepared in Power Query before loading it into the Power BI model. The cleaning workflow included type checking, text trimming/cleaning, capitalization, a conditional Age Group column, and an index column used to support the modeling process.

Queries [3]

healthcare_dataset

Dim_patient

Dim_Admission

✕

✓

fx

Table.AddIndexColumn("#Renamed Columns", "Index", 1, 1, Int64.Type)

Medication

Test Results

Age Group

Patient ID

Index

		100% 0% 0% Valid Error Empty	100% 0% 0% Valid Error Empty	100% 0% 0% Valid Error Empty	100% 0% 0% Valid Error Empty	100% 0% 0% Valid Error Empty
1	tamol	Normal	Young Adult		1	1
2	en	Inconclusive	Middle Age		2	2
3		Normal	Senior		3	3
4	en	Abnormal	Young Adult		4	4
5	in	Abnormal	Adult		5	5
6	en	Normal	Adult		6	6
7	tamol	Inconclusive	Young Adult		7	7
8	tamol	Inconclusive	Young Adult		8	8
9		Abnormal	Senior		9	9
10	tamol	Inconclusive	Middle Age		10	10
11	tamol	Normal	Senior		11	11
12		Inconclusive	Adult		12	12
13	in	Abnormal	Senior		13	13
14		Normal	Senior		14	14
15	tamol	Normal	Adult		15	15
16		Inconclusive	Adult		16	16
17	tamol	Normal	Middle Age		17	17
18	tamol	Abnormal	Adult		18	18
19		Abnormal	Adult		19	19
20	in	Normal	Middle Age		20	20
21		Normal	Senior		21	21
22		Normal	Adult		22	22

Query Settings

PROPERTIES

Name

healthcare_dataset

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Trimmed Text

Cleaned Text

Trimmed Text1

Cleaned Text1

Capitalized Each Word

Added Conditional Column

Added Index

Renamed Columns

Added Index1

Key preparation activities

- Promoted the source row to headers and applied suitable data types.
- Trimmed and cleaned text values to reduce inconsistencies.
- Standardized text using capitalization where required.
- Created an Age Group conditional column for demographic analysis.
- Added an Index column to provide a row-level identifier.

3. Creation of Dim_Patient

The Billing query was merged with the Dim_Patient table using the Patient ID as the matching field. A Left Outer join was used so that all rows from the Billing table were retained while matching patient information was brought in from Dim_Patient.

Merge

Select a table and matching columns to create a merged table.

Billing

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	
Bobby Jackson	30	Male	B-	Cancer	31-01-2024	Matthew Smith	Sons and
Leslie Terry	62	Male	A+	Obesity	20-08-2019	Samantha Davies	Kim Inc
Danny Smith	76	Female	A-	Obesity	22-09-2022	Tiffany Mitchell	Cook PLC
Andrew Watts	28	Female	O+	Diabetes	18-11-2020	Kevin Wells	Hernand

Dim_Patient

Age	Blood Type	Gender	Name	Age Group	Patient ID
30	B-	Male	Bobby Jackson	19-35	1
62	A+	Male	Leslie Terry	60	2
76	A-	Female	Danny Smith	60	3
28	O+	Female	Andrew Watts	19-35	4
43	AB+	Female	Adrienne Bell	36-60	5

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

> Fuzzy matching options

✓ The selection matches 55500 of 55500 rows from the first table.

OK

Cancel

The merge showed that 55,500 of 55,500 rows matched in the first table, confirming that the selected patient key provided the expected match for the displayed merge operation.

4. Creation of Dim_Admission

A second merge was performed between Billing and Dim_Admission. The admission-related attributes were connected using the Admission ID. The Left Outer join preserved all Billing rows and attached matching admission information.

Merge

Select a table and matching columns to create a merged table.

Billing

	Room Number	Admission Type	Discharge Date	Medication	Test Results	Age Group	Patient ID
31	328	Urgent	02-02-2024	Paracetamol	Normal	19-35	1
29	265	Emergency	26-08-2019	Ibuprofen	Inconclusive	60	2
08	205	Emergency	07-10-2022	Aspirin	Normal	60	3
41	450	Elective	18-12-2020	Ibuprofen	Abnormal	19-35	4
01	450	Elective	02-10-2020	Paracetamol	Abnormal	19-35	5

Dim_Admission

1	Doctor	2	Hospital	3	Medical Condition	4	Medication	5	Room Number	6	Test Results	7
	Matthew Smith		Sons and Miller		Cancer		Paracetamol		328		Normal	
	Samantha Davies		Kim Inc		Obesity		Ibuprofen		265		Inconclusive	
	Tiffany Mitchell		Cook PLC		Obesity		Aspirin		205		Normal	
	Kevin Wells		Hernandez Rogers and Vang,		Diabetes		Ibuprofen		450		Abnormal	
	Matthew Smith		Sons and Miller		Cancer		Paracetamol		328		Normal	

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

> Fuzzy matching options

✓ The selection matches 56612 of 56612 rows from the first table.

OK

Cancel

The merge dialog reported 56,612 of 56,612 rows matched from the first table for the displayed operation. This supported the creation of a consolidated Billing query.

5. Final Billing Fact Table

After expanding the merged dimension information, unnecessary columns were removed and the required billing fields were retained. The final Billing query contains the key fields needed for healthcare analysis, including Admission ID, Patient ID, admission and discharge dates, insurance provider and billing-related attributes.

Queries [4]

healthcare_dataset

Dim_Patient

Dim_Admission

Billing

fx

= Table.SelectColumns("#Expanded Dim_Admission",{ "Admission ID", "Patient ID", "Discharge Date",

Admission ID	Patient ID	Discharge Date	Date of Admission	Insurance Provider
Valid 100% 0% 0%	Valid 100% 0% 0%	Valid 100% 0% 0%	Valid 100% 0% 0%	Valid 10
1	1	02-02-2024	31-01-2024	Blue Cross
2	2	26-08-2019	20-08-2019	Medicare
3	3	07-10-2022	22-09-2022	Aetna
4	4	18-12-2020	18-11-2020	Medicare
5	5	09-10-2022	19-09-2022	Aetna
6	50145	09-10-2022	19-09-2022	Aetna
7	5	09-10-2022	19-09-2022	Aetna
8	50145	09-10-2022	19-09-2022	Aetna
9	6	24-12-2023	20-12-2023	UnitedHealthcare
10	7	15-11-2020	03-11-2020	Medicare
11	8	07-01-2022	28-12-2021	Cigna
12	9	14-07-2020	01-07-2020	Cigna
13	10	22-06-2021	23-05-2021	UnitedHealthcare
14	11	22-04-2020	19-04-2020	Medicare
15	12	05-09-2023	13-08-2023	Medicare
16	13	28-12-2019	12-12-2019	Cigna
17	14	19-06-2020	22-05-2020	UnitedHealthcare
18	15	13-10-2021	08-10-2021	UnitedHealthcare
19	16	11-01-2023	01-01-2023	Blue Cross
20	17	14-07-2020	23-06-2020	UnitedHealthcare
21	18	02-04-2020	08-03-2020	Cigna
22	19	14-03-2021	04-03-2021	Blue Cross
23	20	08-11-2023	08-11-2023	Blue Cross

Query Settings

PROPERTIES

Name

Billing

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Trimmed Text

Cleaned Text

Trimmed Text1

Cleaned Text1

Capitalized Each Word

Added Conditional Column

Merged Queries

Expanded Dim_Patient

Merged Queries1

Expanded Dim_Admission

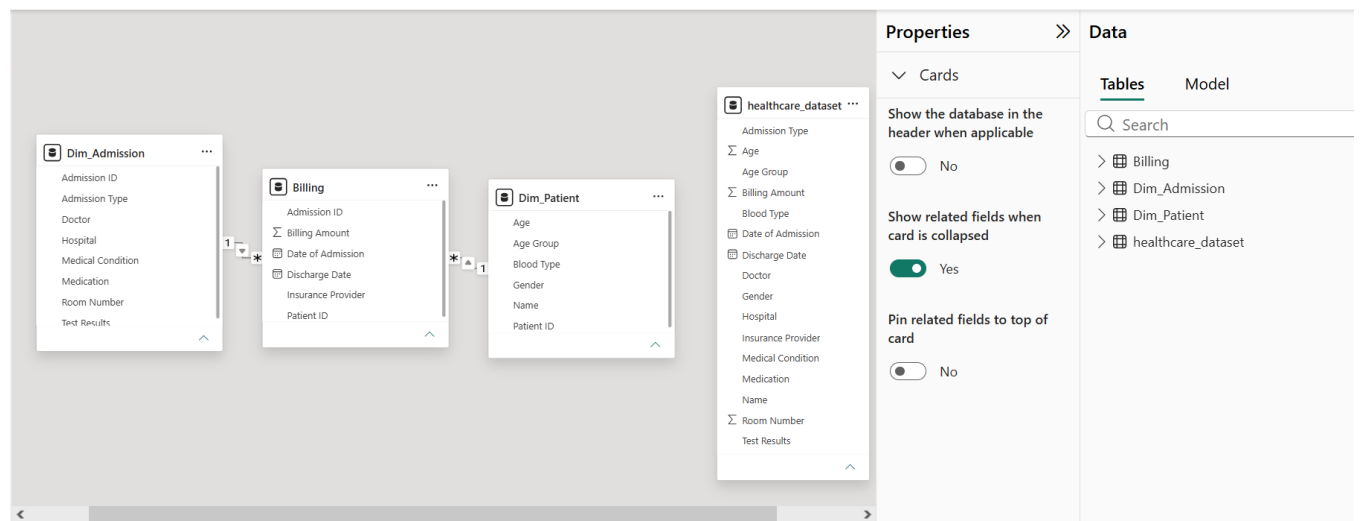
Removed Other Columns

Fact-table role

Billing acts as the central fact table in the model. It stores the event-level information used for measures and visual analysis, while Dim_Patient and Dim_Admission provide descriptive attributes.

6. Data Model and Relationships

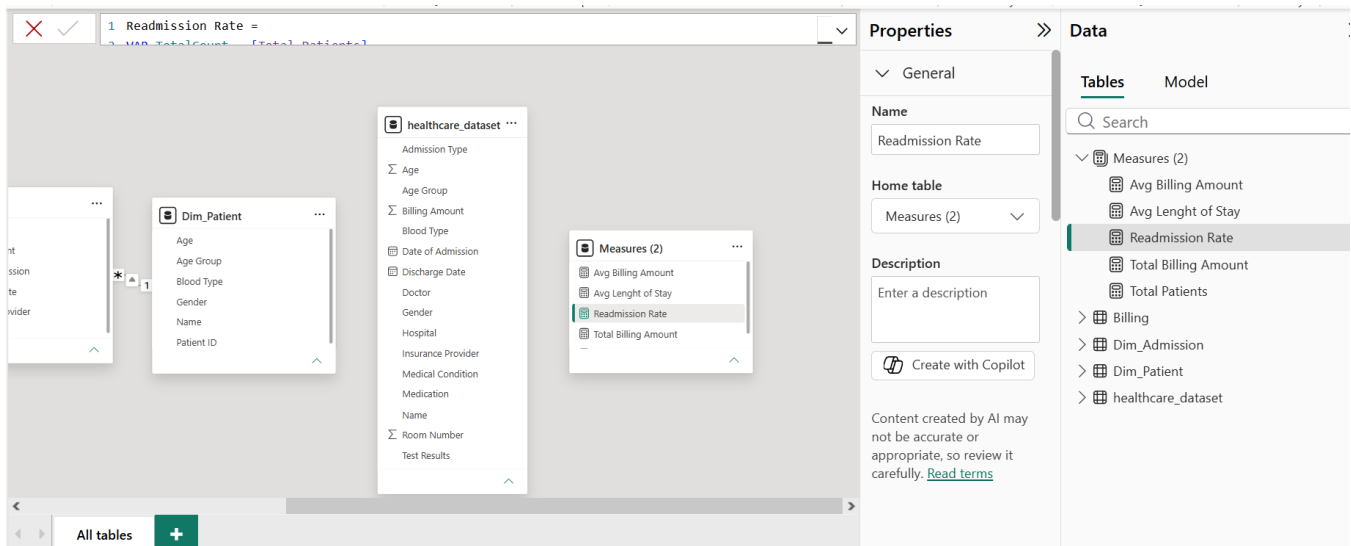
The cleaned tables were loaded into the Power BI model and arranged into a star-schema style structure. Billing is connected to the patient and admission dimensions through their corresponding keys.



The model enables patient-level and admission-level attributes to filter billing measures and dashboard visuals. This structure supports reusable DAX calculations and interactive filtering.

7. DAX Measures

Measures were created in a dedicated Measures table to support the dashboard. The report includes measures for total patients, total billing amount, average billing amount, average length of stay and readmission rate.

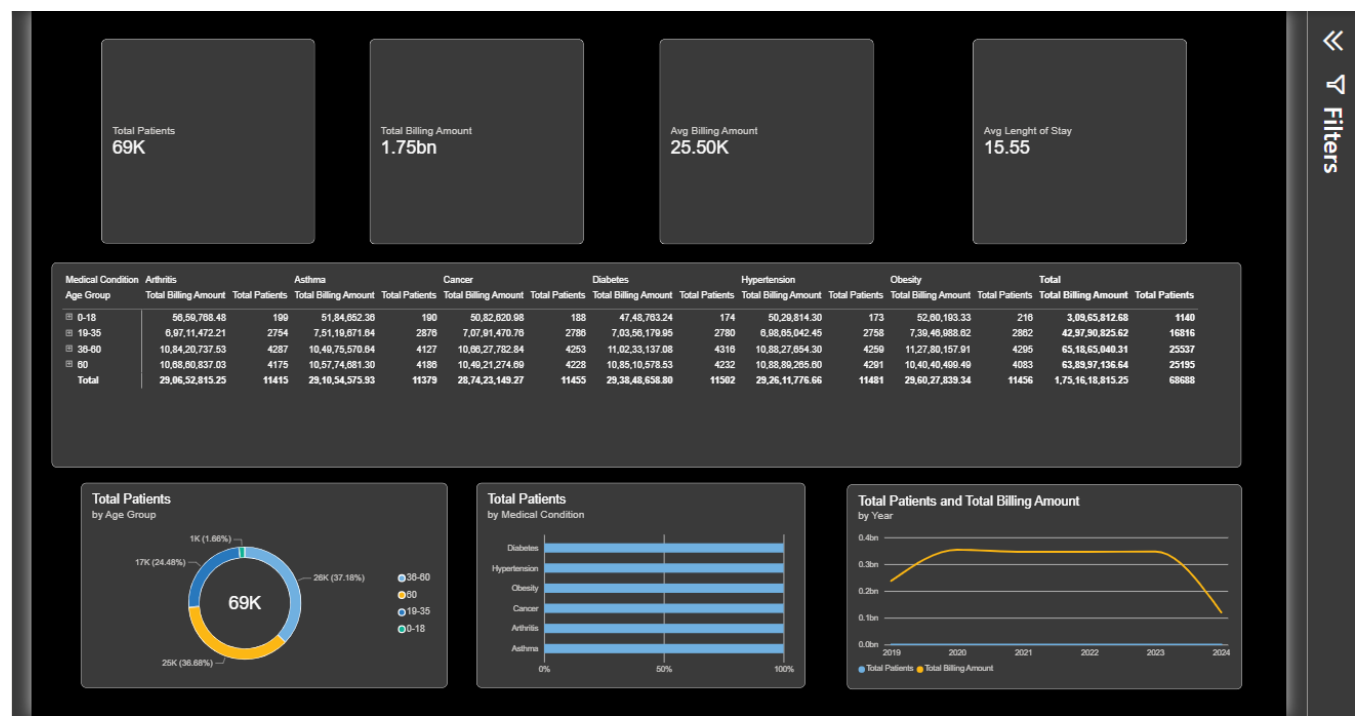


Measures used

- Total Patients – summarizes the patient count used in the report.
- Total Billing Amount – summarizes the overall billing amount.
- Avg Billing Amount – calculates the average billing amount.
- Avg Length of Stay – summarizes the average patient stay.
- Readmission Rate – provides a percentage-based indicator for readmission analysis.

8. Interactive Power BI Dashboard

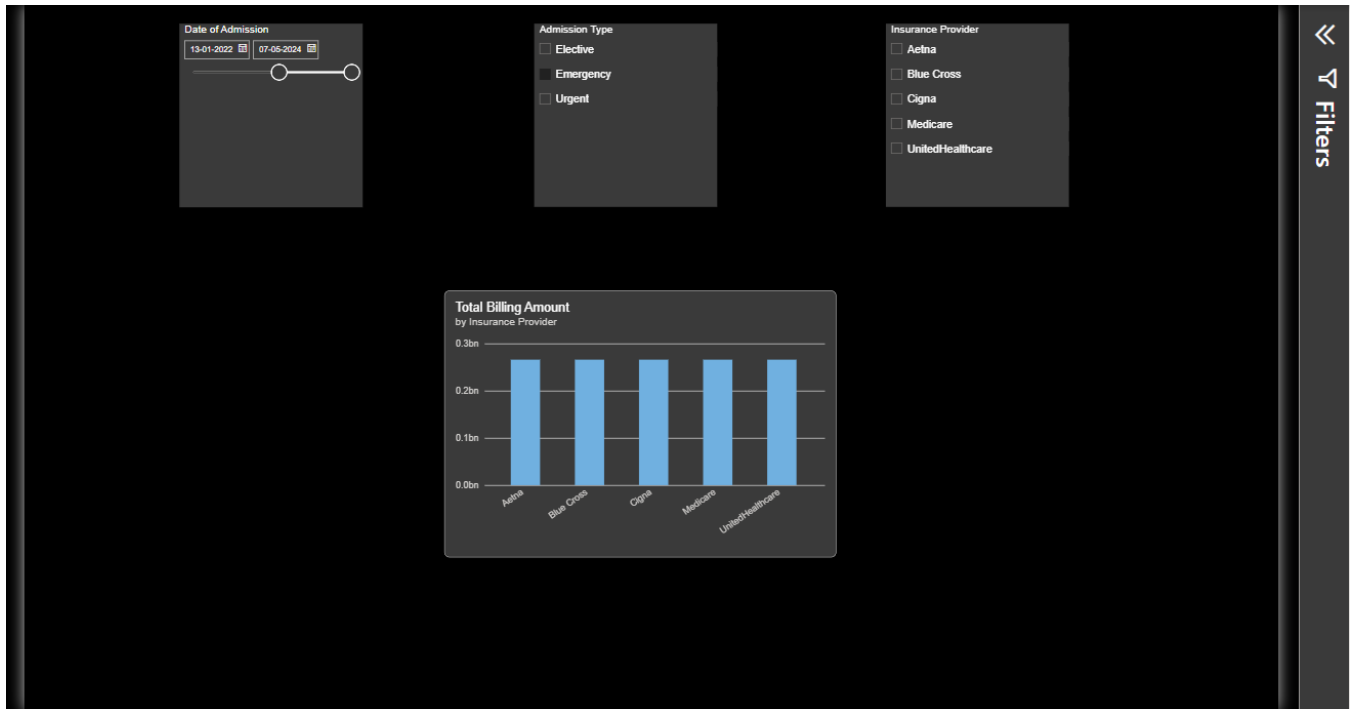
The final dashboard combines KPI cards, a matrix, a donut chart, a bar chart and a line chart to present healthcare information in an easy-to-read format.



The dashboard provides a high-level view of patient volume, billing amount and average length of stay. It also compares patient distribution across age groups and medical conditions and shows year-wise trends.

9. Interactive Filters and Detailed Analysis

Slicers were added to make the dashboard interactive. Users can filter the report by admission date, admission type and insurance provider. The billing amount chart updates according to the selected filter context.



Interactive features

- Date range slicer for Admission Date.
- Admission Type slicer for Emergency, Elective and Urgent cases.
- Insurance Provider slicer for provider-level comparison.
- Cross-filtering of visuals based on the selected slicer values.

10. Conclusion

The Week 5 Healthcare Power BI project was completed by transforming the raw healthcare data into a cleaner and more structured analytical model. Power Query was used for data preparation and standardization, including Age Group classification and text cleaning. Billing was organized as the central fact table, while Dim_Patient and Dim_Admission were used as supporting dimension tables.

The final Power BI model supports DAX-based measures and interactive visuals. The completed dashboard provides a clear view of patient counts, billing amounts, average length of stay, medical-condition distribution, age-group distribution, year-wise trends and insurance-provider analysis. Slicers make the report interactive and allow users to explore the healthcare data from different perspectives.

Skills Demonstrated

- Power Query data cleaning and transformation
- Conditional column and index creation
- Merge and expand operations
- Dimension and fact table design
- Power BI data modeling and relationships
- DAX measure creation
- Interactive dashboard design and filtering